Matrix Theory II–III: Rank and Systems of Linear Equations: Answer Key

Problem sheet for the Day 2 classes · CSIR NET / IIT JAM previous year questions

Answers

Q	1	2	3	4	5	6	7	8	9	10
Answer	7	(D)	(A)	(A)	(B)	(D)	(D)	(B),(D)	(A),(D)	(A),(C)

Hints and short solutions

1. **7** JAM 2019 $R_2 - R_1 : (0,1,2,3 \mid 1)$, $R_3 - R_1 : (0,2,4,k-1 \mid 1)$, then $R_3 - 2R_2 : (0,0,0,k-7 \mid -1)$. Inconsistent iff $k = 7$.
2. **(D)** CSIR NET Dec 2016 Each block row has rank 1 ($\operatorname{rank} J = 1$) and the three blocks sit in different rows and columns: $1+1+1 = 3$.
3. **(A)** CSIR NET Dec 2016 Every column sums to 0, so $(1,\dots,1)A = 0$: the rows are dependent and the rank is at most $n-1$.
4. **(A)** CSIR NET Dec 2015 $A^tAx = 0 \Rightarrow \|Ax\|^2 = 0$, so $\ker A^tA = \ker A$ and $\operatorname{rank} A^tA = \operatorname{rank} A$.
5. **(B)** CSIR NET June 2016 Uniqueness means $\ker A = 0$, so $\operatorname{rank} A = m \le n$.
6. **(D)** CSIR NET June 2019 Subtract each row from the next: the differences $(0,1,1,1,1), (0,0,1,1,1), \dots$ give a triangular matrix with non-zero diagonal.
7. **(D)** CSIR NET June 2018 BA is $n \times n$ of rank at most $m < n$. AB ($m \times m$) may or may not be singular.
8. **(B),(D)** CSIR NET Dec 2015 $A(\lambda u + (1-\lambda)v) = \lambda b + (1-\lambda)b = b$ for every λ. The solution set misses 0, so it is not a subspace. Rank n means $\ker A = 0$.
9. **(A),(D)** CSIR NET Dec 2017 Solvable for every b means the column space is $\mathbb{R}^m$, so $r = m \le n$. If also $m \ge n$, then $m = n$ and A is invertible (so (C) fails).
10. **(A),(C)** CSIR NET Dec 2014 $4 = \operatorname{rank} AB \le \operatorname{rank} A, \operatorname{rank} B \le 4$. So B (with 4 columns) has nullity 0, and $\operatorname{rank} BA \le 4 < 7$.